PDUFA and Controlled Substance Approvals

Progress Report

ECON 580 Senior Thesis — 3rd Status Update

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Research Question

Core question:

Did the Prescription Drug User Fee Act (PDUFA, 1992) increase the share of FDA-approved drugs that are DEA-scheduled controlled substances?

Motivation:

- PDUFA introduced industry-funded user fees that expanded FDA review capacity and increased approval throughput
- Faster reviews $\Rightarrow$ more drugs approved sooner — but did this disproportionately benefit drugs with **abuse potential**?
- Conceptual channel: regulatory acceleration $\rightarrow$ greater CS supply $\rightarrow$ downstream diversion and misuse risk

This presentation: descriptive findings, first-pass event study results, honest accounting of problems, and paths forward.

Roadmap

1. What PDUFA is — and what it is not
2. Data construction and sample
3. Descriptive results (8 figures)
4. Event study results (5 figures)
5. Problems and limitations (6 issues)
6. Where the thesis stands
7. Paths forward and discussion

What PDUFA Does

- Enacted October 1992; reauthorized every 5 years (now PDUFA VII, through 2027)
- Authorizes FDA to collect fees from firms submitting drug applications
- Fees fund **FDA staffing** to review applications faster — not specific approval decisions
- **Three fee types:**
  - Application fee (≈$4M for a full NDA with clinical data)
  - Annual establishment fee
  - Annual product fee
- Fees are **uniform** — same rate regardless of drug type, therapeutic area, or schedule status
- **Key point:** fees are not scaled to profitability, revenue potential, or abuse/diversion risk

What User Fees Are NOT (1/2)

Does the FDA retain all decision-making authority regarding the marketing authorization of medical products?

Yes. For products where premarket review is required, there are standards for FDA product review that are written into law and that the agency follows regardless of the source of funding. The FDA's work, high standards, and decisions are guided by science and focused on protecting and promoting the public health.

Importantly, there is no direct connection between a fee paid to submit an application and the review outcome for that application. The fees are not a "fee-for-service" payment. Rather, the user fees, including application fees, are authorized by Congress to help the FDA fund payroll and related costs to deliver on program goals.

User fees provide the FDA with supplemental resources for reviewing product applications based on the scientific and clinical evidence, and for conducting related activities.

Source: FDA PDUFA program documentation

The FDA is explicit: user fees are **not "fee-for-service"** payments. They fund payroll and review capacity, not specific approval decisions.

What User Fees Are NOT (2/2)

Importantly, the outcomes of decisions the FDA makes (such as whether to approve a product) do not depend on the agency's ability to collect user fees. The FDA's decisions are made based on science and are consistent with the legal/regulatory standards that govern the agency.

Source: FDA PDUFA program documentation

"The outcomes of decisions the FDA makes (such as whether to approve a product) do not depend on the agency's ability to collect user fees."

Critical Institutional Detail: ANDAs Not Covered Until 2012

- PDUFA (1992) applied to **NDAs and BLAs only** — new drugs and biologics
- Generic drug applications (ANDAs) got their own user fee program, **GDUFA**, only in **2012**
- **This matters enormously because:**
  - 86.6% of controlled substance approvals in our sample are **ANDAs**
  - These were not subject to PDUFA user fees for the first 20 years
  - Top CS sponsors are all generic manufacturers
- **Implication:** the "user fees create incentives" channel does **not** operate through the pathway where most CS action occurs
- If anything, GDUFA (2012) is the more relevant policy shock for the generics-as-CS story

Data Sources

- **FDA: Drugs@FDA** — all approved drug applications (NDA, ANDA, BLA), downloaded March 2026
  - 12 relational tables; submission events keyed by `ApplNo + SubmissionType + SubmissionNo`
  - Covers approved applications only (failed/withdrawn not observed)
- **DEA: Controlled Substance Schedules** — official scheduling classifications (Schedules I–V), March 2026
  - Linked at the **active ingredient** level
  - If any ingredient matches a DEA-scheduled substance, the drug is flagged
  - **Current schedules, not historical** — a limitation we will return to
- **Conservative definition:** only confident, exact DEA schedule matches count as controlled substances

Sample Construction

Step	Observations	Dropped
Full FDA backbone (all submission events)	191,266	—
Keep original approved (ORIG + AP)	26,268	164,998
Deduplicate by application (earliest date)	26,076	192
Drop 2026 partial year + null dates	25,908	168
Final sample	**25,908**	—

- One row per unique approved drug application (`ApplNo`)
- Year range: 1939–2025
- Unit: the **first original approval** for each application

Sample Overview

By application type:

Type	N
ANDA (generic)	19,022 (73.4%)
NDA (new drug)	5,277 (20.4%)
BLA (biologic)	460 (1.8%)
Other/unknown	1,149 (4.4%)

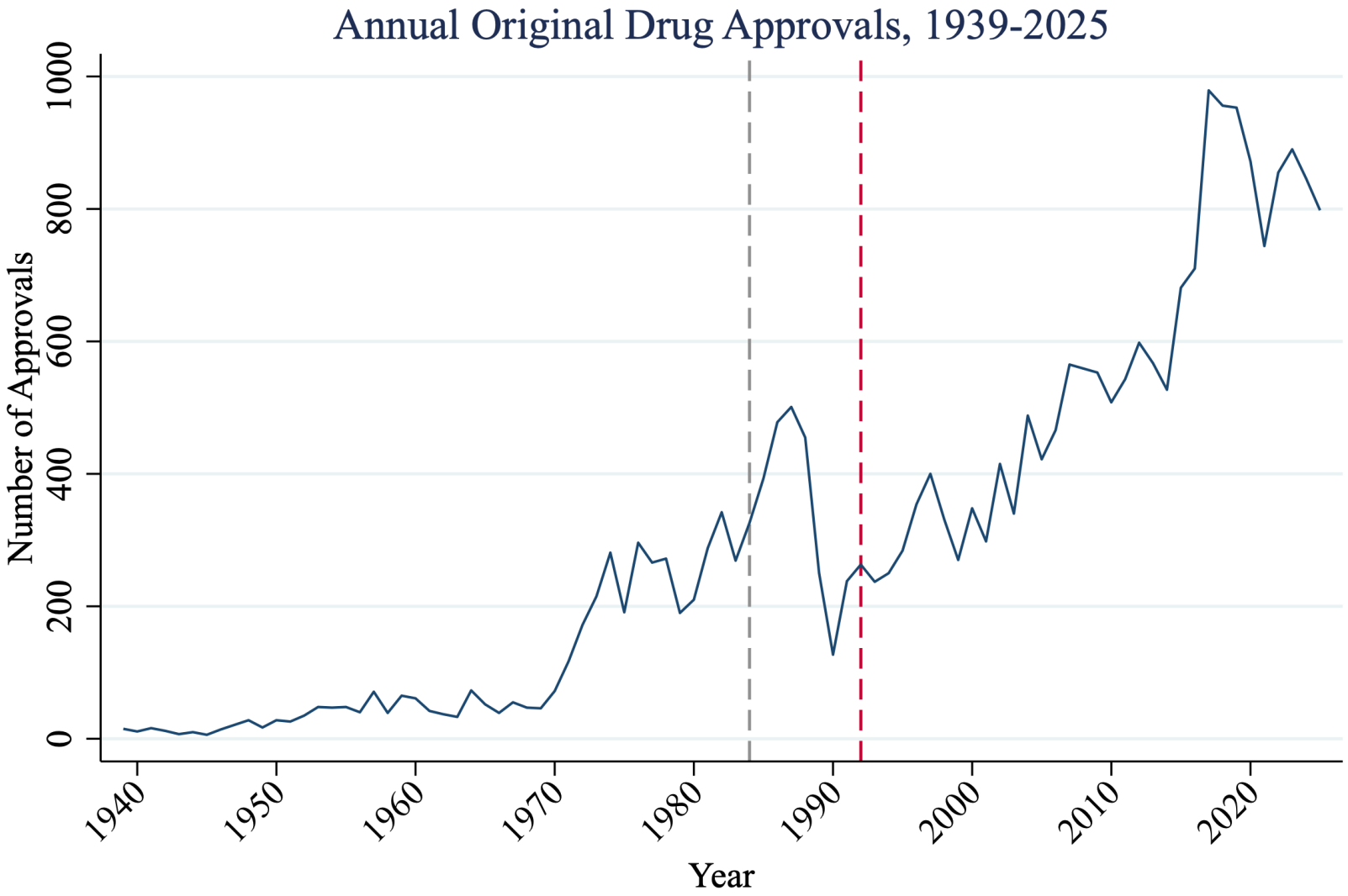
Controlled substances (CS = 2,067, 8.0%):

Schedule	N
II	994 (48.1%)
IV	769 (37.2%)
III	189 (9.1%)
V	108 (5.2%)
I	7 (0.3%)

Key Fact: Controlled Substances Are Overwhelmingly Generics

- Among 2,067 CS approvals:
  - **1,790 are ANDAs (86.6%)**
  - Only **276 are NDAs (13.4%)**
- CS share *within* type:
  - Among ANDAs: 9.4% are CS
  - Among NDAs: 5.2% are CS
- Top 20 sponsors of CS approvals are **all generic manufacturers** (Watson, Hikma, Sandoz, Barr, Qualitest, Mallinckrodt, . . .)
- No major innovator pharmaceutical companies appear in the top 20
- **What this tells us:** the CS approval story is a **generic drug entry story**, not a novel-drug-development story
- That is a **Hatch-Waxman (1984) story**, not a PDUFA story

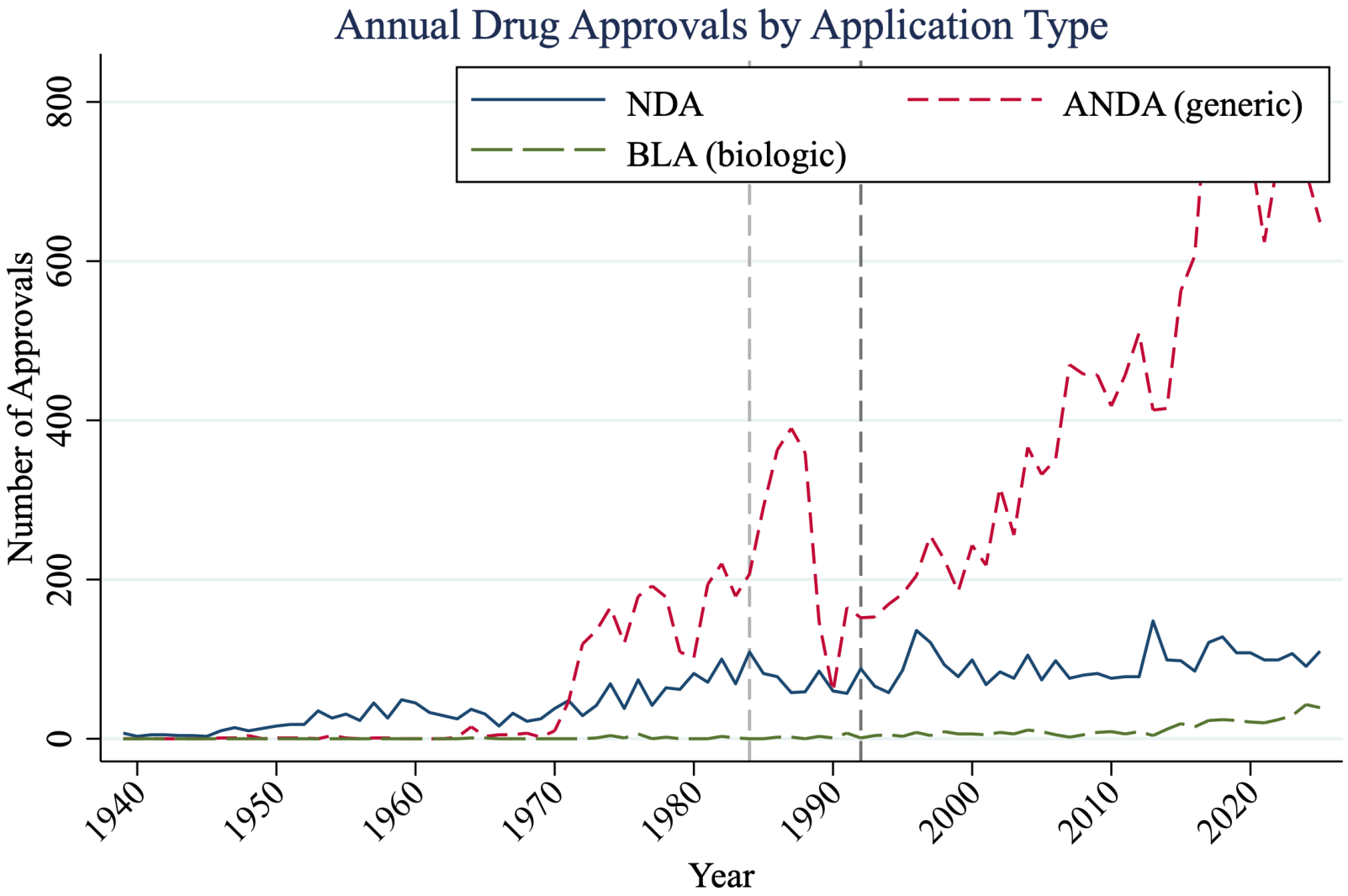
Total Drug Approvals Over Time



Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Massive growth in total approvals, especially after Hatch-Waxman (1984) — the generic-drug entry mechanism. This is the denominator

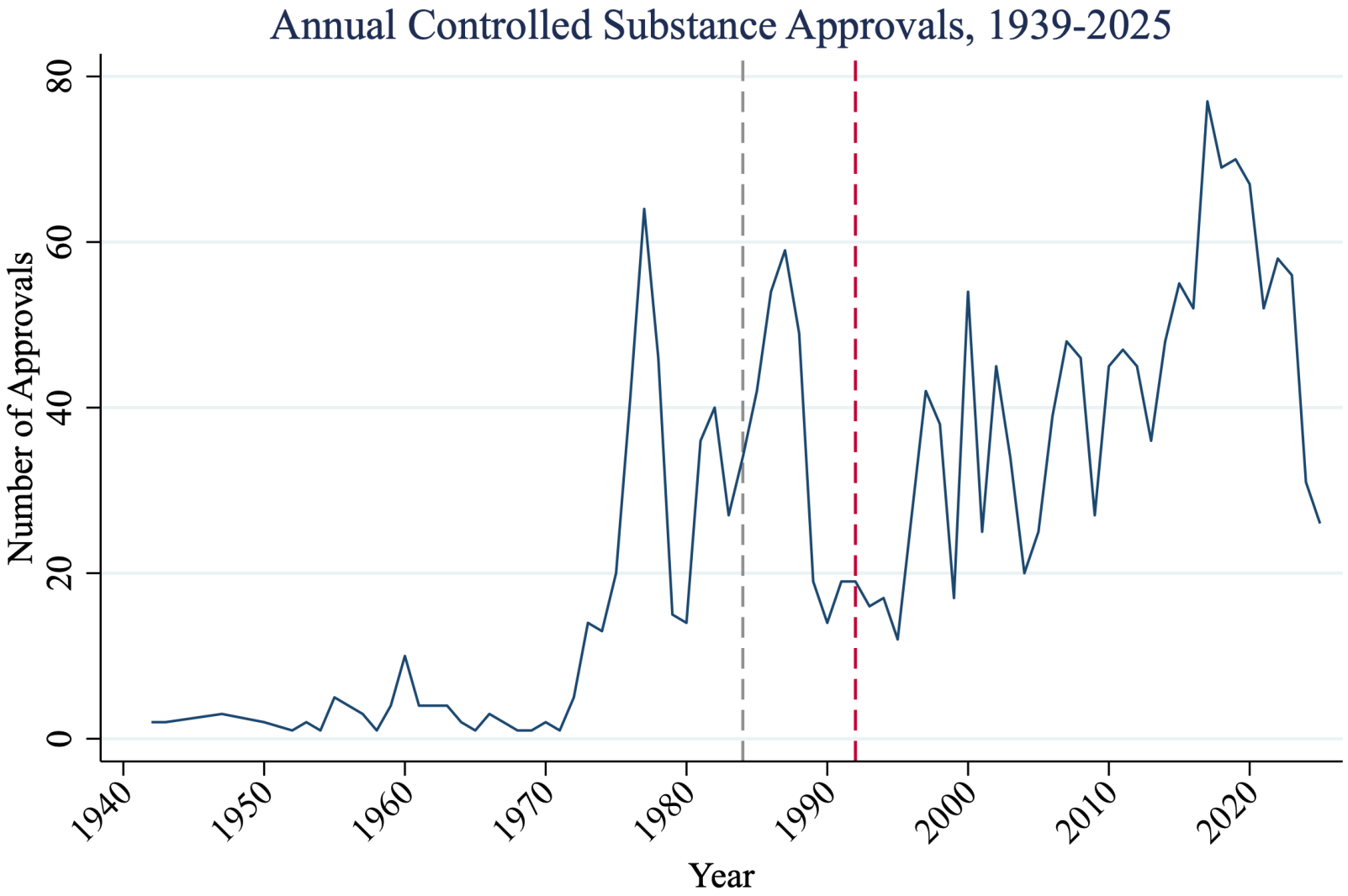
Approvals by Application Type



Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Growth is almost entirely ANDA-driven. NDA approvals are relatively flat. BLAs appear only after 1982. The composition of the approval

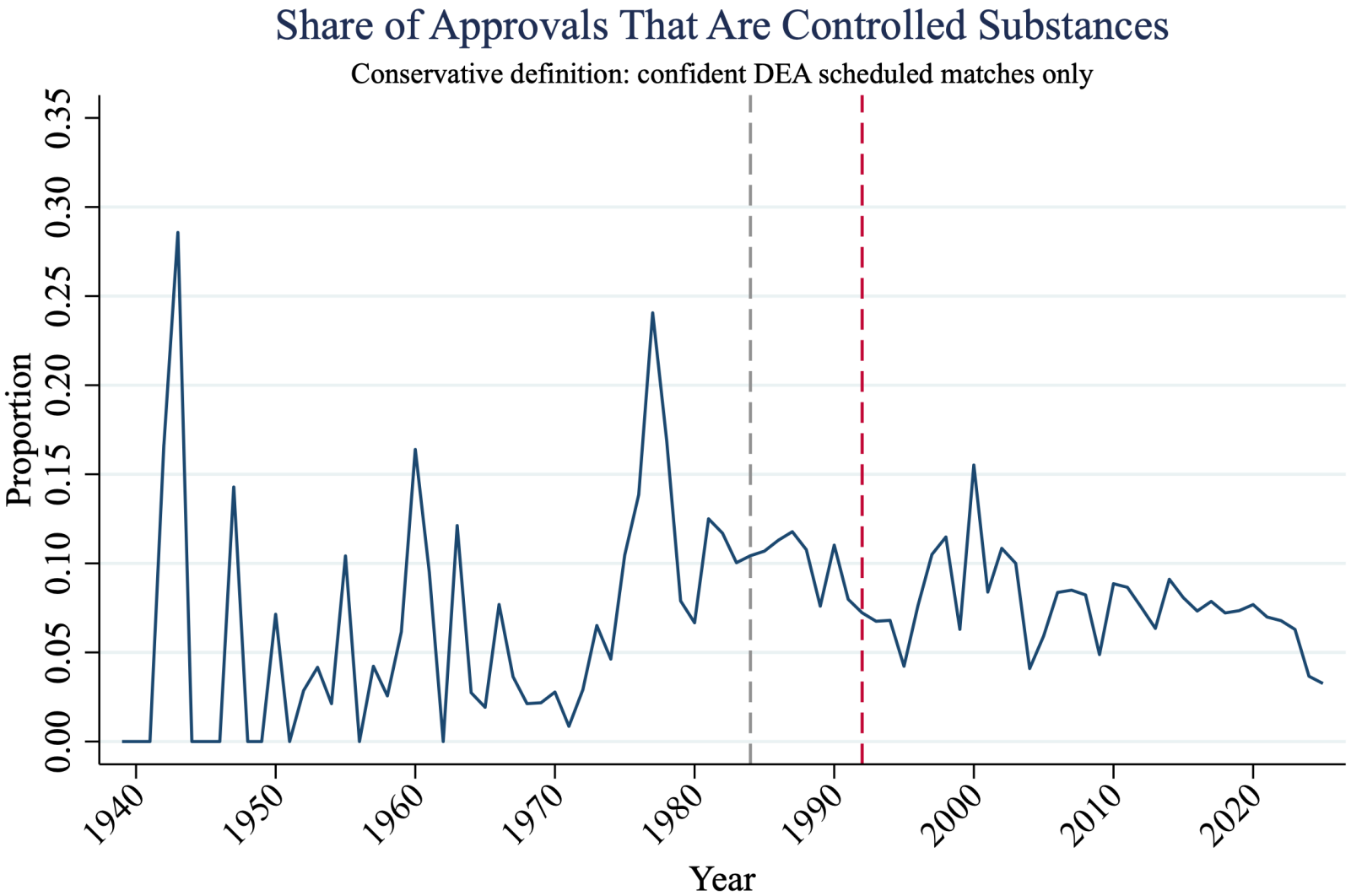
Annual Controlled Substance Approvals (Count)



Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

CS approvals grew in absolute terms — from low single digits pre-1970 to 40–70/year recently. But total approvals grew even faster.

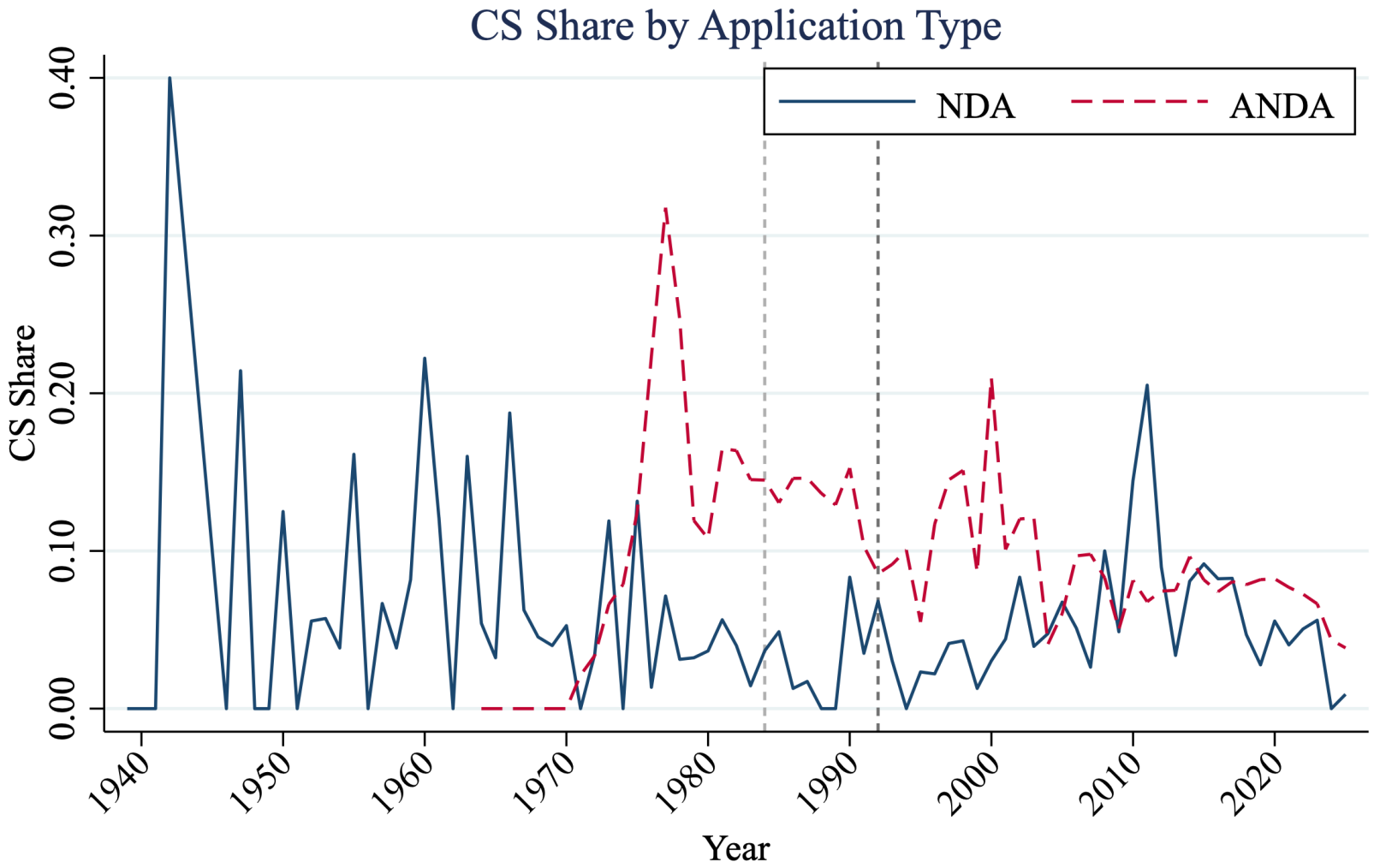
Share of Approvals That Are Controlled Substances



Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Main descriptive figure. CS share was highest in the 1940s–1970s. It has been on a general decline since the early 1980s. No visible upward

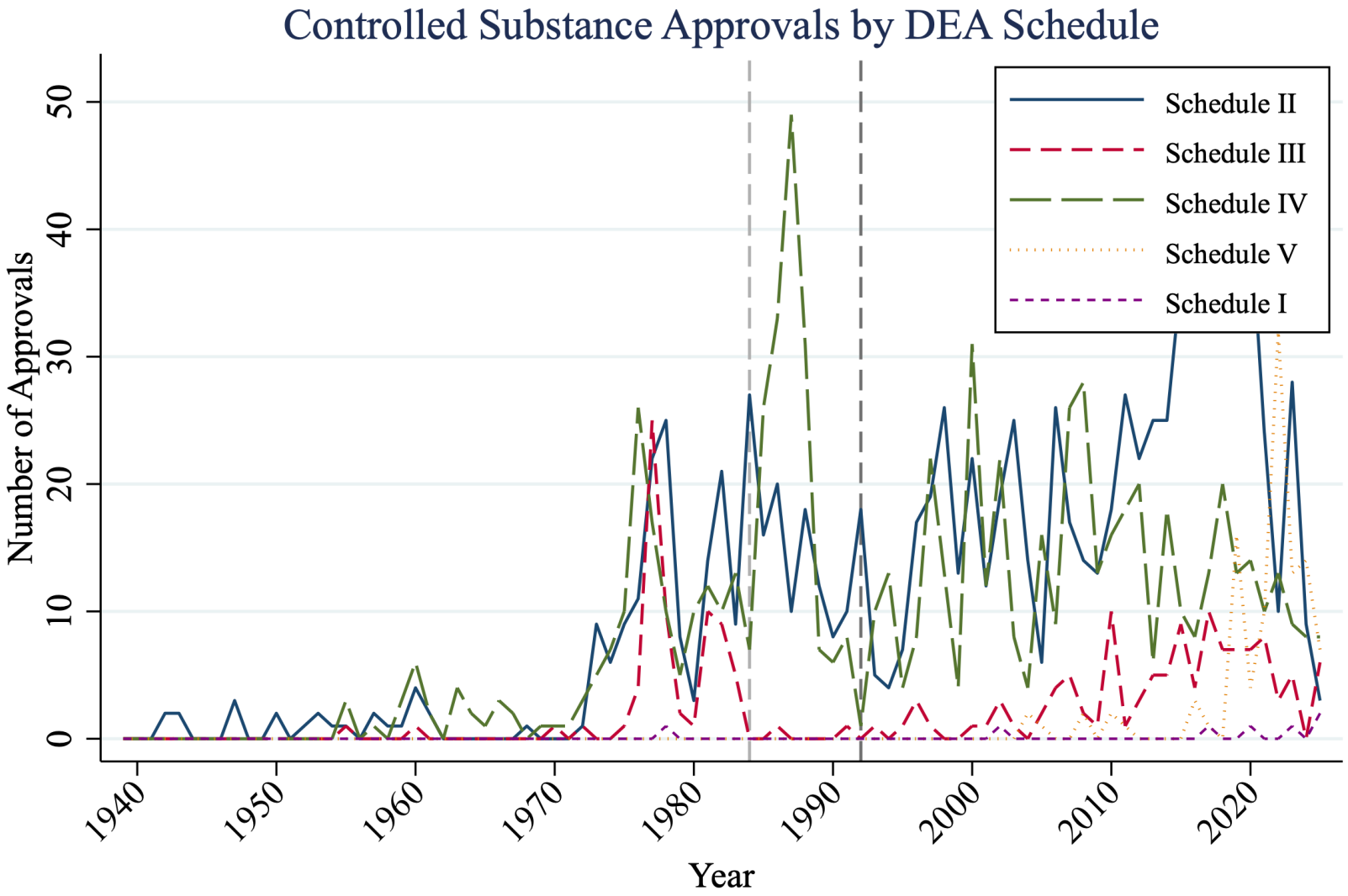
CS Share by Application Type (NDA vs. ANDA)



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Years with < 5 NDA or ANDA approvals excluded for that type.
Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

NDA CS share is volatile but generally higher than ANDA CS share. ANDA CS share spikes in early 1980s (Hatch-Waxman era) then settles

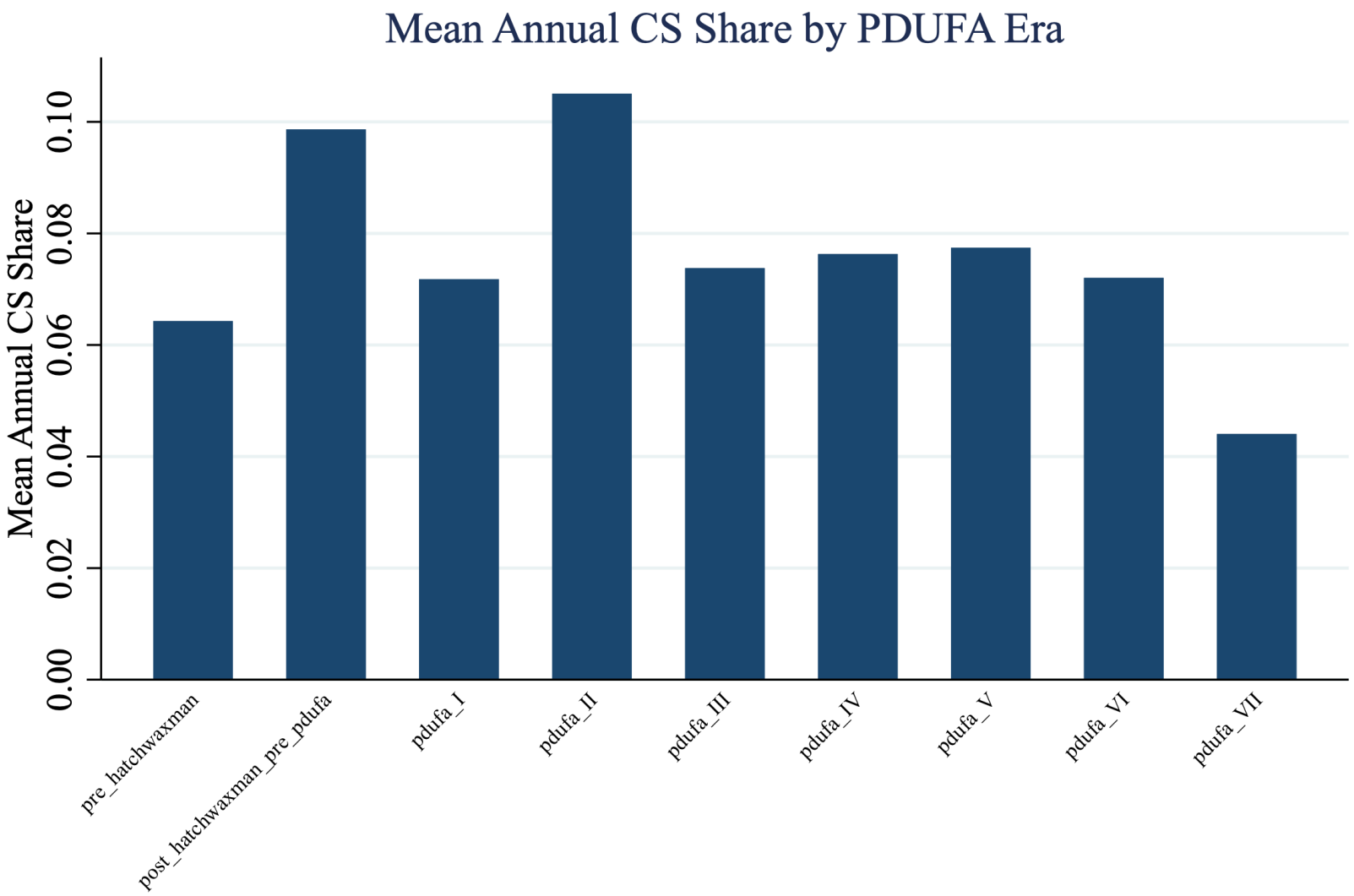
Controlled Substance Approvals by DEA Schedule



Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Schedule II and IV dominate. Schedule II (opioids, stimulants) has grown notably since ≈2005. The growth in Schedule II generics drives

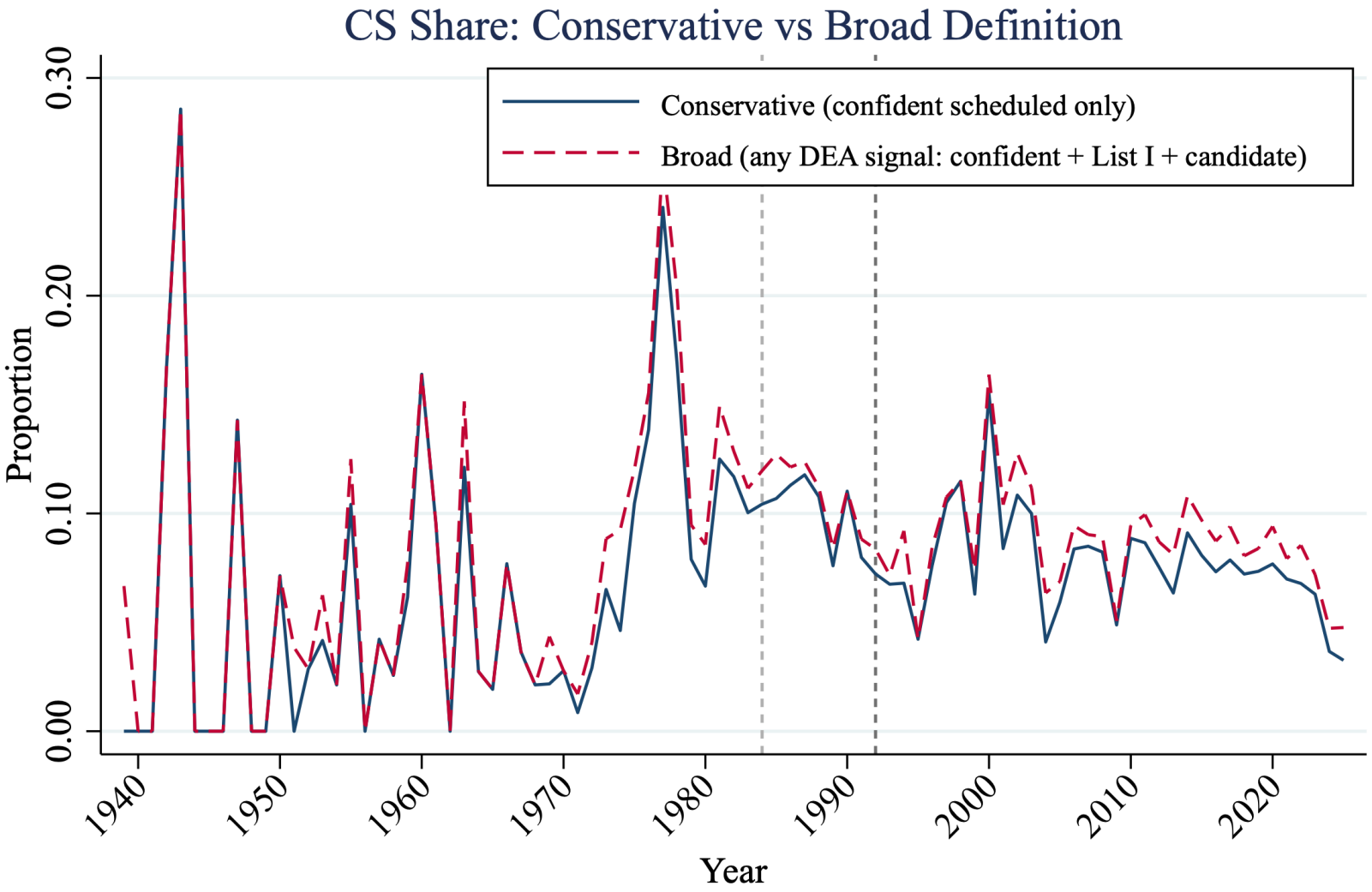
Mean CS Share by PDUFA Era



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

CS = Controlled Substance. No monotonic increase across PDUFA eras. PDUFA VII (2023+) has the lowest mean CS share. PDUFA II had

Conservative vs. Broad CS Definition



CS = Controlled Substance.
Dashed lines: 1984 (Hatch-Waxman) and 1992 (PDUFA).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

The two definitions track closely. The conservative measure (confident DEA matches only) is not systematically missing a large set of

Pre-PDUFA vs. Post-PDUFA Summary

Pre-PDUFA (1939–1992):

- Avg. 135 approvals/year
- Avg. ≈13 CS approvals/year
- CS share: ≈9.6%

Post-PDUFA (1993–2025):

- Avg. 564 approvals/year
- Avg. ≈41 CS approvals/year
- CS share: ≈7.3%

Bottom line

CS approvals tripled in raw count. CS *share* fell. Total approvals quadrupled. The ANDA explosion inflated the denominator faster than the numerator.

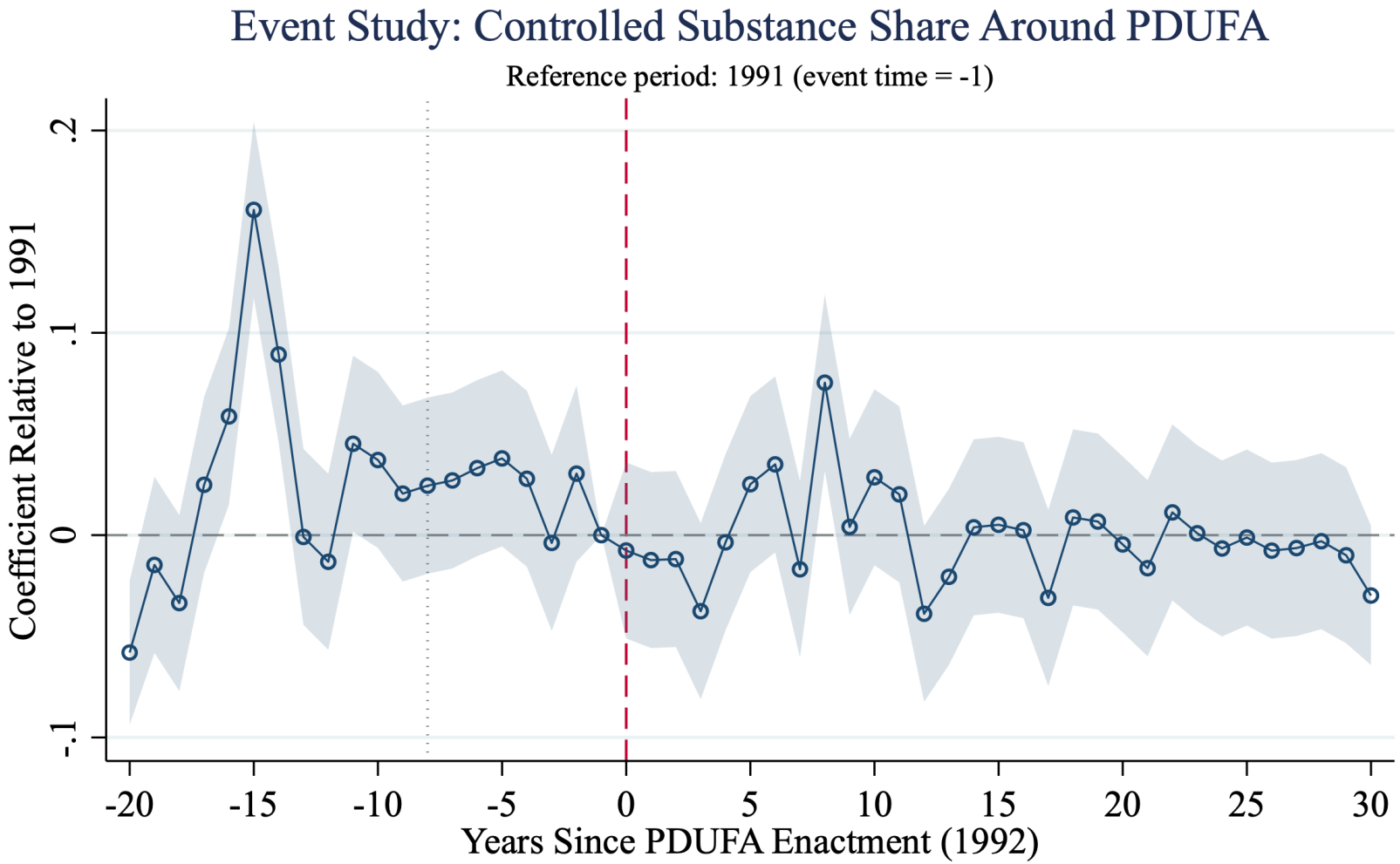
Event Study Design

- **Design:** interrupted time series (ITS) centered on PDUFA enactment (1992)
- Annual data, **1970–2025** (pre-1970 dropped: tiny denominators, very volatile)
- **Reference year:** 1991 (event time $= -1$)
- **End-caps:** event time ≤ -20 and $\geq +30$ binned
- **No cross-sectional variation** — single national time series
- **Cannot** identify a causal treatment effect; no untreated units
- Results are descriptive policy diagnostics, not causal estimates

Three outcome measures:

1. CS share of total approvals (main)
2. Log CS approval counts
3. NDA-only CS share (cleanest conceptual test)

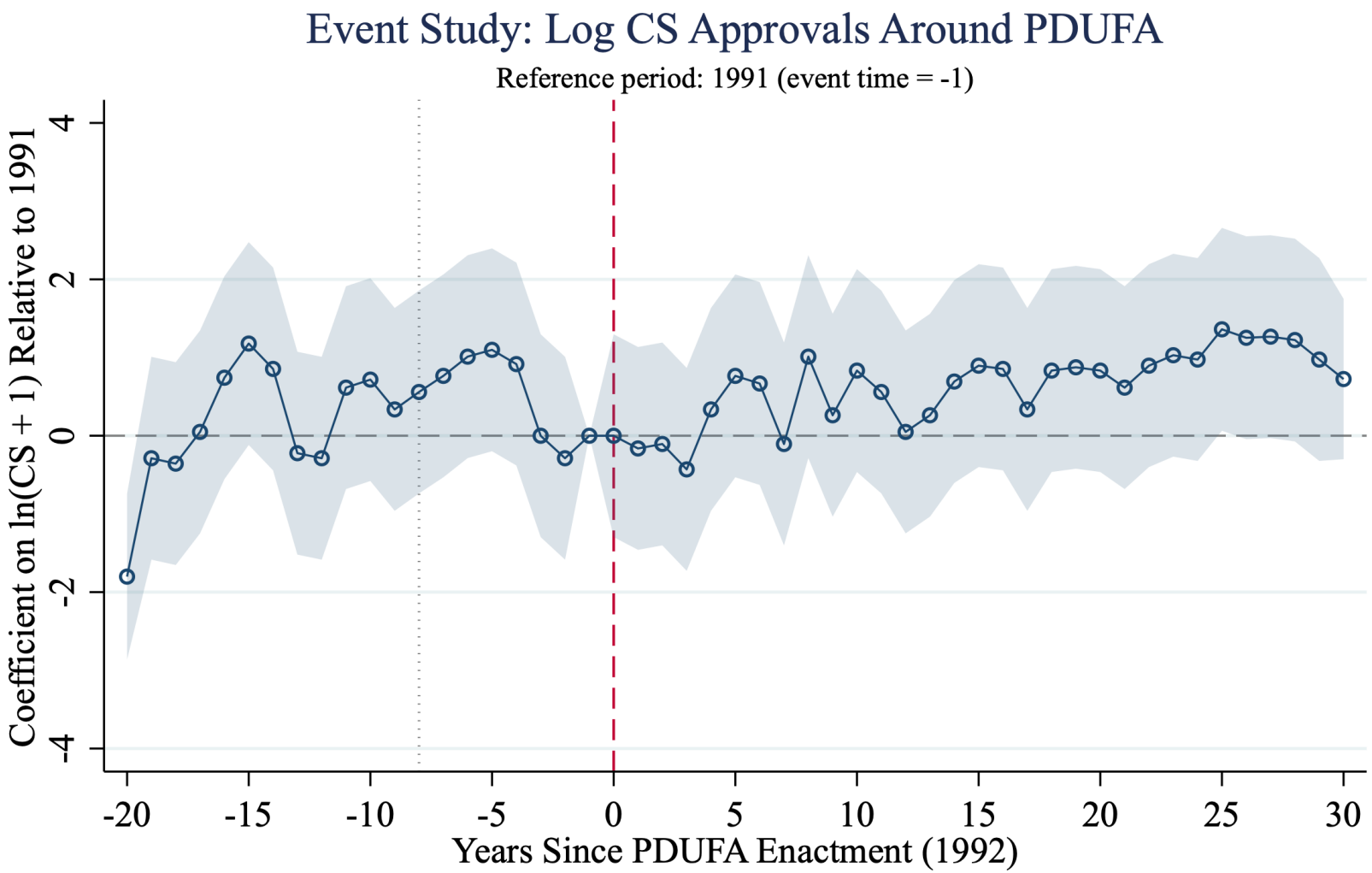
Event Study — CS Share (Main Result)



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Dashed red line: PDUFA enactment (event time = 0, 1992).
Dotted line: Hatch-Waxman Act (event time = -8, 1984).
Shaded band: 95% CI. Interrupted time series. No cross-sectional variation. Treat as descriptive diagnostics.
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

No visible break at event time 0. Pre-event coefficients are not flat (spike at ≈ -15, likely reflecting 1970s scheduling activity). Post-event

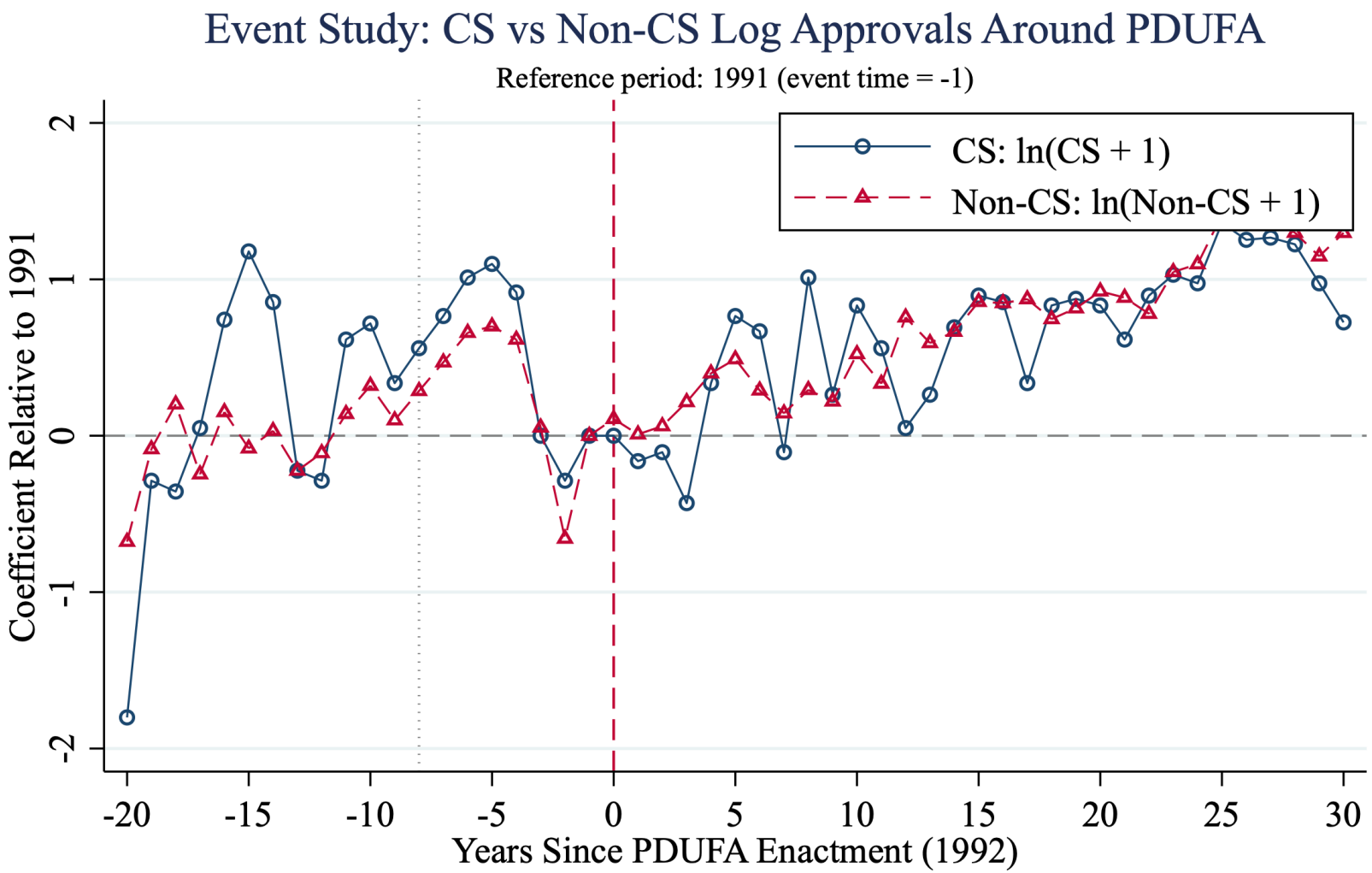
Event Study — Log CS Counts



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Dashed red line: PDUFA enactment. Dotted line: Hatch-Waxman (1984).
Shaded band: 95% CI. Interrupted time series. No cross-sectional variation. Treat as descriptive diagnostics.
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

CS = Controlled Substance. Log CS approvals trend upward over time, but growth is gradual with no sharp break at 1992. Wide confidence

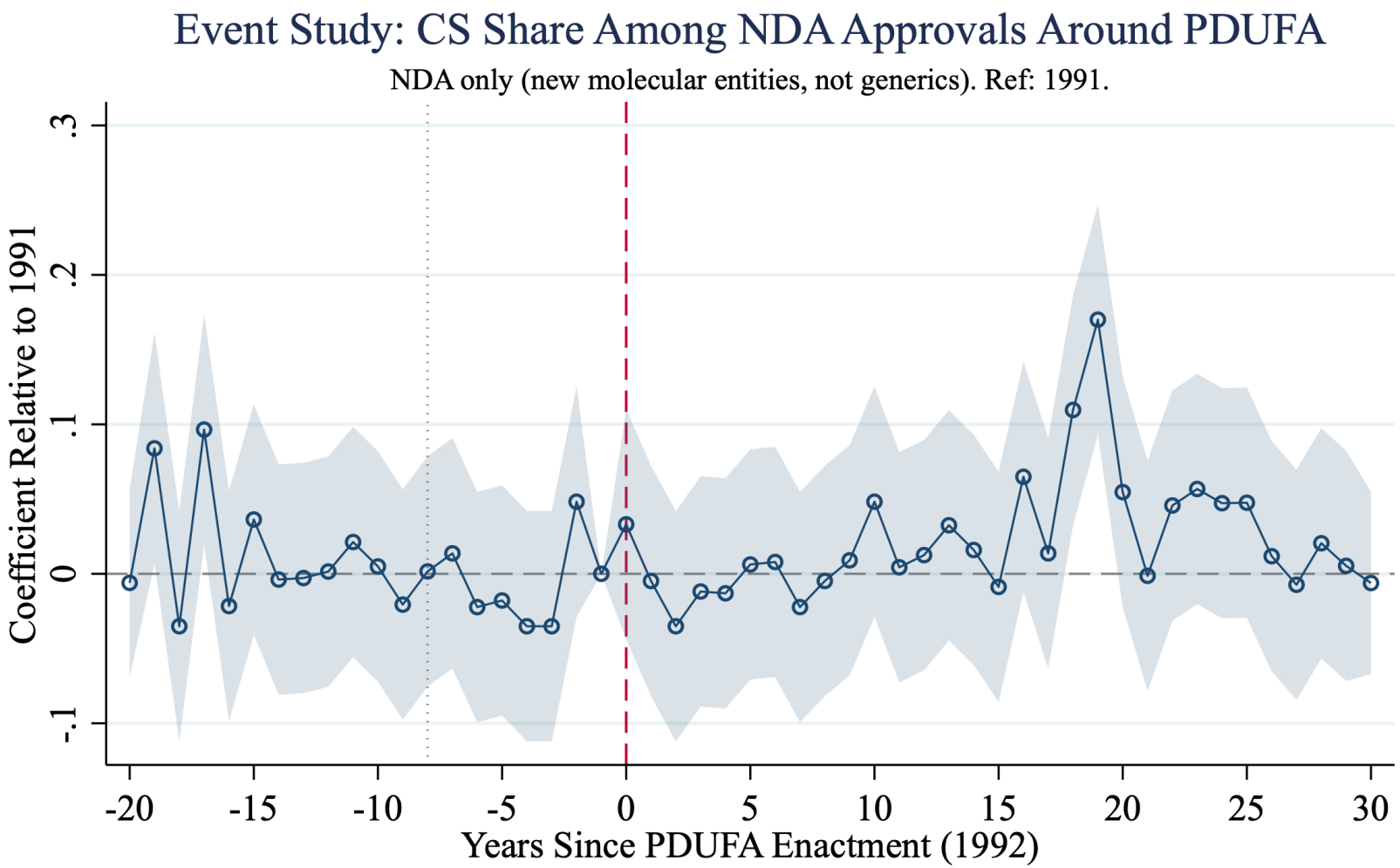
Event Study — CS vs. Non-CS Log Counts



CS = Controlled Substance. Non-CS = all other approvals.
Dashed red line: PDUFA enactment. Dotted line: Hatch-Waxman (1984).
Interrupted time series. No cross-sectional variation. Treat as descriptive diagnostics.
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Both CS and non-CS approvals grow after PDUFA and track each other through the 1990s–2000s. Non-CS pulls ahead after ≈2005, driving

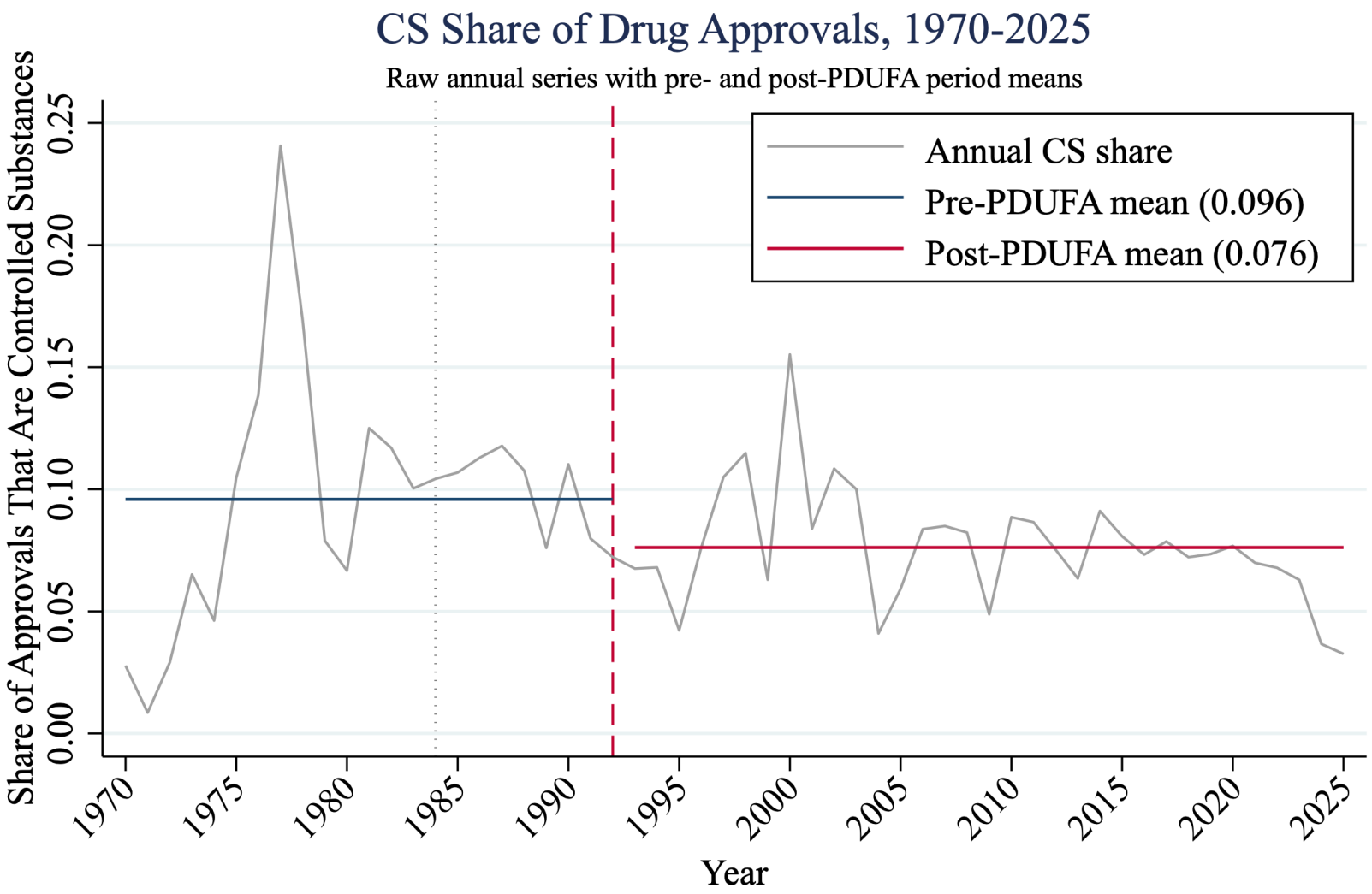
Event Study — NDA-Only CS Share



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Restricted to NDA applications. Small annual denominators — wide CIs expected.
Dashed red line: PDUFA enactment. Dotted line: Hatch-Waxman (1984).
Shaded band: 95% CI. Interrupted time series. No cross-sectional variation. Treat as descriptive diagnostics.
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

Confidence intervals span 30+ percentage points. Pure noise. Sample too small (≈30–50 NDAs/year) to detect anything meaningful.

Summary Figure: Raw Series with Pre/Post Means



CS = Controlled Substance (conservative: confident DEA scheduled matches only).
Dashed red line: PDUFA enactment (1992). Dotted line: Hatch-Waxman (1984).
Source: FDA Drugs@FDA, DEA controlled substance schedules. ORIG+AP submissions only.

CS = Controlled Substance. Raw CS share with pre-PDUFA mean (≈9.6%, 1970–1992) and post-PDUFA mean (≈7.3%, 1993–2025). The

Parametric Results: Did the CS Share Shift at PDUFA?

Annual panel, 56 observations (1970–2025). Outcome: CS share of total approvals.

Model A — Simple level shift:

$$\text{CS Share}_t = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{Post-PDUFA}_t] + \varepsilon_t$$

	Coef.	Std. Err.	p-value	R^2
Post-PDUFA (β_1)	-0.020	0.010	0.052	
Constant (β_0)	0.096	0.008	0.000	0.068

The average CS share was 9.6% before PDUFA and about 2 pp lower after. Marginally significant ($p = 0.052$). The CS share fell — it did not rise.

Parametric Results: Controlling for Trend and Slope Change

Model B — Level shift + linear time trend:

$$\text{CS Share}_t = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{Post}_t] + \beta_2 \cdot \text{Year}_t + \varepsilon_t$$

	Coef.	Std. Err.	p-value
Post-PDUFA (β_1)	-0.015	0.019	0.446
Year (β_2)	-0.0002	0.0006	0.759

$R^2 = 0.070$. $N = 56$.

Model C — Level shift + trend + slope change:

$$\text{CS Share}_t = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{Post}_t] + \beta_2 \cdot \text{Year}_t + \beta_3 \cdot (t - 1992) \times \mathbf{1}[\text{Post}_t] + \varepsilon_t$$

	Coef.	Std. Err.	p-value
Post-PDUFA (β_1)	-0.025	0.019	0.205
Year (β_2)	0.002	0.001	0.128
Slope change (β_3)	-0.003	0.001	0.054

$R^2 = 0.135$. $N = 56$.

Once we control for trend, the PDUFA shift is statistically zero. The slope change ($\beta_3 = -0.003$, $p = 0.054$) suggests a marginally significant downward acceleration — the **opposite** *direction of the hypothesis.*

Parametric Results: Log Counts

Model D — Log CS approvals, trend + slope change:

$$\ln(\text{CS}_t + 1) = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{Post}_t] + \beta_2 \cdot \text{Year}_t + \beta_3 \cdot (t - 1992) \times \mathbf{1}[\text{Post}_t] + \varepsilon_t$$

	Coef.	Std. Err.	p-value
Post-PDUFA (β_1)	-0.704	0.311	0.028
Year (β_2)	0.070	0.018	0.000
Slope change (β_3)	-0.040	0.021	0.064

$R^2 = 0.409$. $N = 56$.

CS approvals grow over time ($\beta_2 > 0$, $p < 0.001$). But the post-PDUFA level shift is negative ($\beta_1 = -0.704$, $p = 0.028$) — CS counts were below trend right after PDUFA. The slope change is also negative: CS approval growth slowed after 1992.

Bottom line

Across all four models: no evidence that PDUFA increased controlled substance approvals, whether measured as a share or in log counts.

Problem 1: DEA Linkage Is Retroactive

- We apply **current (2026) DEA scheduling** to drugs approved decades ago
- A drug approved in 1975 may not have been scheduled at approval — or may have had a different schedule
- Scheduling status changes: drugs get rescheduled up or down (e.g., hydrocodone combination products moved II $\rightarrow$ III $\rightarrow$ II)
- We have **no historical scheduling data**
- **Consequence:** the level of the CS share in any given year may be mismeasured; changes over time may partly reflect DEA rescheduling events rather than changes in approval composition
- Pre-event spike ($\approx$event time -15, i.e. $\approx$1977) may reflect a historical scheduling wave that is now retroactively coded into our sample

Problem 2: The Denominator Problem Is Fundamental

- Total approvals quadrupled after PDUFA, driven by the ANDA explosion
- Even if CS approvals grew, the *share* falls if non-CS approvals grow faster
- The share-based outcome **conflates numerator and denominator dynamics**
- Log counts partially address this, but do not cleanly separate "more CS drugs" from "more drugs overall"
- The ideal approach: model CS and non-CS entry separately; test for *differential* growth rates
- This is not merely a scaling issue — it is a question about what the economically meaningful outcome measure is
- **Candidate reframing:** study the absolute supply of controlled substances (level) rather than their share of approvals

Problem 3: Sample Mixes New Molecular Entities with Generics

- An ANDA for generic oxycodone $\neq$ an NDA for a novel opioid
- The PDUFA-as-incentive hypothesis applies to **NDAs** (new drug development decisions by innovator firms)
- But **86.6%** of CS approvals are ANDAs — generic copies of existing controlled substances
- NDA-only analysis is conceptually cleaner but statistically underpowered ($\approx$30–50 NDAs/year; CIs span 30 pp)
- **What the data actually describe:** a **generic-drug proliferation story**, not an innovation story
- More Schedule II and IV generic copies of existing brand-name controlled substances entering the market after Hatch-Waxman
- That is a **Hatch-Waxman story**, not a PDUFA story

Problem 4: Hatch-Waxman (1984) Confounds the Pre-Period

- Hatch-Waxman created the entire ANDA pathway for generics in 1984
- This is arguably the **larger structural change** in the approval landscape — bigger than PDUFA
- The pre-PDUFA period (1970–1992) *includes* this massive shock 8 years before our treatment
- Pre-event coefficients in the event study are **not flat** — there is a visible pre-trend that likely reflects Hatch-Waxman dynamics
- We cannot cleanly separate Hatch-Waxman effects from PDUFA effects with this design
- **The two policies are bundled in calendar time** and both affect the composition of the approval pool
- One mitigation: restrict to NDAs only; but then we lose power

Problem 5: No Correction for Serial Correlation

- Annual CS share is almost certainly **autocorrelated** — drug approvals have multi-year regulatory lags, and scheduling waves affect multiple years at once
- OLS standard errors do not account for this; CIs may be too narrow
- Newey-West (HAC) standard errors with lag ≈ 3 would be appropriate for a paper-facing version
- With only $\approx$55 annual observations, HAC corrections have limited small-sample reliability in any case
- **Qualitative conclusion is likely unchanged** (most point estimates are small or near zero regardless), but formal inference is imprecise
- This is noted throughout the do-file but not yet corrected in the current output

Problem 6: Single National Time Series $\neq$ Causal Design

- **No untreated units.** No cross-sectional variation in treatment timing. Cannot construct a counterfactual.
- This is an **interrupted time series** — the weakest quasi-experimental design
- Any contemporaneous shock is confounded with PDUFA:
  - Changes in DEA scheduling practice in the 1990s
  - Pharmaceutical R&D portfolio shifts
  - Opioid crisis and related regulatory responses
  - Economic cycles affecting generic entry timing
- Miller (2023, JEP): with only treated units and a common event date, treatment effect identification requires very strong counterfactual assumptions
- **Bottom line:** even if we observe a post-1992 change, we cannot attribute it to PDUFA with any confidence

What We Can Honestly Conclude

- PDUFA did **not** increase the *share* of FDA-approved drugs that are controlled substances — the share declined
- Raw *counts* of CS approvals tripled (from ≈13 to ≈41/year), but this growth was proportional to or slower than overall approval growth
- The CS approval increase is concentrated in **Schedule II generics** filed through the ANDA pathway
- Top sponsors are all generic manufacturers; no innovator firms in the top 20
- **No evidence that PDUFA created differential incentives for controlled substance development**
- The data describe a Hatch-Waxman + generic proliferation story, not a PDUFA story

Revised Framing of the Question

Original framing

"Did PDUFA's user fee structure incentivize more controlled substance approvals?"

Problems with this framing:

- User fees are uniform — not drug-type-specific
- ANDAs (86.6% of CS approvals) weren't covered by PDUFA until 2012
- The composition margin shows no positive shift at 1992

Weaker but potentially defensible framing

"Did PDUFA-era regulatory acceleration, by increasing overall FDA throughput, lead to greater *absolute supply* of controlled substances — even without changing the composition?"

The data suggest even this is better attributed to Hatch-Waxman than PDUFA.

Potential Paths Forward

1. **Shift from composition to supply:** 41 CS approvals/year vs. 13 may matter for downstream outcomes (diversion, misuse) even if the share didn't change
2. **Study GDUFA (2012):** a natural experiment specifically targeting generic drug user fees — the pathway that dominates CS approvals
3. **Fix the DEA linkage:** obtain historical scheduling data (DEA Orange Book archives, Federal Register notices)
4. **Link to downstream outcomes:** ARCOS (retail distribution), CDC WONDER (overdose mortality), NFLIS (drug seizure data)
5. **Narrow the question:** focus on Schedule II opioids or stimulants specifically, not all CS
6. **Intensive margin:** study *review time* for CS vs. non-CS drugs under PDUFA, rather than approval composition
7. **Pivot the question entirely** — discuss

Open Questions for Discussion

1. Is the **null result on composition** itself a publishable/interesting finding? ("PDUFA didn't change CS share because the mechanism doesn't apply to generics")
2. Should the thesis pivot to studying **absolute supply growth** rather than composition?
3. Is **GDUFA (2012)** a more tractable policy shock? What identification challenges does it bring?
4. What downstream data would make the "supply $\rightarrow$ misuse" connection credible? Is ARCOS accessible?
5. Is there a better identification strategy that could address the Hatch-Waxman confound?
6. Should the thesis **pivot the question entirely**? If so, toward what? Opioid prescribing? Generic entry timing? Something else entirely?

Thank You

Thank you

Questions and feedback welcome

Summary

- Data built, DEA linkage built, descriptive results complete
- First-pass event study complete
- Core hypothesis (PDUFA $\rightarrow$ more CS approvals as share) not supported by the data
- The CS story is a generics/Hatch-Waxman story, not a PDUFA story
- Many open questions — feedback on paths forward most useful

All code, data, and figures available in the thesis repository at `/Users/alexdelatorre/Desktop/econ580-thesis`.